CASE STUDY REPORT

Real-World Virus and Cyber Attacks on Systems, Corporations and Government Bodies

Five Cases: Timing, Causes, Losses, Recovery and Solutions

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Introduction and Overview

This report studies five well-documented cyber attacks in which malware, spyware or wiper software caused serious damage to servers, corporations, government bodies or digital data. Three cases are from around the world and two are from India. For each case the report describes when it happened, what caused it, what was lost, how the victim recovered, what it cost, and what solutions were adopted.

No.	Case	Year	Region	Attack Type
1	Target data breach	2013	USA	POS malware, card theft
2	Sony Pictures hack	2014	USA	Wiper malware, data leak
3	Bangladesh Bank heist	2016	Bangladesh	Malware, SWIFT fraud
4	Kudankulam nuclear plant	2019	India	DTrack spyware
5	Air India data breach	2021	India	Vendor attack, data theft

Note on Sources

The facts in this report are taken from widely reported public sources such as government statements, news agencies and company reports. Figures such as losses are estimates and may differ slightly between sources. Where details were not officially confirmed, the report says so. Screenshots of the original articles should be placed in the marked boxes with the article headline, publication name and date as a caption.

Case 1: Target Corporation Data Breach (USA, 2013)

Date / Timing	Attack ran from 27 November to 15 December 2013 (Thanksgiving to Christmas shopping season); discovered 12 December 2013 after the US Department of Justice alerted the company; made public 19 December 2013
Victim	Target Corporation, second-largest discount retailer in the USA
Attack type	Point-of-sale (POS) malware, a RAM scraper known as BlackPOS, installed on checkout machines
Data stolen	About 40 million credit and debit card records plus personal details of about 70 million customers

What happened

Attackers planted malware on Target's checkout terminals in thousands of stores. It copied card data from the memory of the machines as customers swiped their cards, and sent it to servers controlled by the criminals. The stolen cards were then sold on underground markets.

Cause

- Attackers first stole network login credentials of Fazio Mechanical, an HVAC (heating and cooling) contractor of Target, through a phishing email.
- Target's payment network was not properly separated from the vendor-accessible network.
- Security software reportedly raised alerts about the malware, but these were not acted on in time.

Losses

- Target reported gross expenses of about $162 million related to the breach (about $105 million after insurance and tax effects).
- Holiday-season sales and profit dropped and customer trust fell.
- In 2017 Target agreed to an $18.5 million settlement with US states; banks and card companies also claimed compensation.
- The CEO, Gregg Steinhafel, resigned in May 2014.

Recovery and Cost

Target removed the malware, offered a free year of credit monitoring to affected customers, and reissued compromised cards through banks. The company invested about $100 million to roll out chip-and-PIN card readers in all its stores.

Solution / Lessons

- Separate the payment network from other networks and vendor access.
- Strict control and monitoring of third-party vendor accounts.
- Act on security alerts quickly.
- Use chip-and-PIN (EMV) card technology.

Source Evidence (Article / Photo)

- Suggested search: "Target data breach December 2013 40 million cards Reuters / BBC / New York Times"
- Other sources: US Senate Commerce Committee report "A Kill Chain Analysis of the 2013 Target Data Breach" (March 2014), Krebs on Security



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THE DAILY CHRONICLE

TRUTH INFORMS PROGRESS

VOL. 142, NO. 318 | THURSDAY, DECEMBER 19, 2013 | RIVERDALE, USA | 75 CENTS

TARGET HIT BY MASSIVE CYBER BREACH: 40 MILLION CREDIT CARDS COMPROMISED

Attack Traced to HVAC Vendor Credentials During Holiday Rush

By Megan Ellis
Staff Writer

MINNEAPOLIS — Target Corp. confirmed Wednesday that cybercriminals stole credit and debit card information from as many as 40 million customers in a massive data breach that struck during the height of the holiday shopping season.

The retailer said the attackers gained access to its systems using credentials from a third-party vendor that provides heating, ventilation and air conditioning (HVAC) services, underscoring how even trusted suppliers can become an unintentional gateway for hackers.

"This was a sophisticated and targeted attack," said Gregg Steinhafel, Target's chairman, president and chief executive officer, at a news conference in Minneapolis. "We are deeply sorry for the concern and frustration this has caused our guests."

The breach, which occurred between November 27 and December 15, affected card transactions made at Target stores across the United States. The company said encrypted PIN data does not appear to have been compromised.

Federal investigators, including the FBI and the U.S. Secret Service, hd

Experts Say Breach Highlights Growing Risks for Retailers

By Samantha Greene
Technology Correspondent

Security experts say the Target breach is a stark reminder of how vulnerable even the largest retailers can be in an era of increasingly sophisticated cyber attacks.

"This wasn't a simple hack," said James McQuade, a cybersecurity analyst at the Robertson Institute. "Attackers leveraged real vendor credentials, which shows a deep understanding of how companies operate. It's a wake-up call for every business that works with third parties."

Analysts say the breach could have far-reaching consequences for the retail giant, including potential lawsuits, regulatory scrutiny and lasting damage to customer trust.

"Target will need to invest heavily in security, transparency and customer support to move past this," McQuade said. "The question now is how quickly they can rebuild confidence." See ANALYSIS, Page A8

Shoppers Express Anger and Concern

By David R. Coleman
Staff Writer

MINNEAPOLIS — For many holiday shoppers, the news of Target's data breach landed like a punch to the gut.

"I trusted Target. It's where I shop for my family, for everything," said Angela Morris, a mother of two from St. Paul, while leaving a Target store Wednesday. "It's scary to think my information could be out there."

Long lines were still evident at many locations, but some customers said they planned to use cash or other forms of payment until they felt confident their data was safe.

"The timing makes it worse," said college student Derrick James. "People are already stretched thin during the holidays, and now we have to worry about fraud too."

See SHOPPERS, Page A6

Case 2: Sony Pictures Entertainment Hack (USA, 2014)

Date / Timing	24 November 2014; leaks continued through December 2014; US government named North Korea on 19 December 2014
Victim	Sony Pictures Entertainment, Culver City, California
Attacker	Group calling itself "Guardians of Peace"; the FBI attributed the attack to North Korea
Attack type	Destructive wiper malware plus mass data theft

What happened

Employees saw a red skeleton image on their screens and found their computers locked or wiped. The hackers leaked unreleased Sony films, private emails of executives, salary details and personal data (including social security numbers) of about 47,000 current and former employees. The attack came before the release of the comedy film "The Interview", which mocked North Korea's leader.

Cause

- Believed to have started with spear-phishing emails to employees.
- Reports said Sony had weak internal security practices, including passwords stored in plain files and poor monitoring of large data transfers.

Losses

- Sony said it spent about $15 million on investigation and recovery in the quarter after the attack; later estimates of total cost go up to around $35 million.
- Thousands of computers and servers were wiped.
- Leaked emails damaged reputations and led to the exit of senior executives such as co-chairperson Amy Pascal.
- Sony paid a class-action settlement to employees (reported at up to about $8 million).

Recovery and Cost

For weeks employees used pen and paper, fax machines and old phones. Sony hired the security firm Mandiant to investigate, rebuilt its network, replaced thousands of machines and provided identity-theft protection to employees. Core systems were restored within a few weeks.

Solution / Lessons

- Encrypt and monitor sensitive data.
- Segment internal networks and limit admin privileges.
- Train staff against phishing.
- Have a crisis communication and disaster recovery plan.

Source Evidence (Article / Photo)

- Suggested search: "Sony Pictures hack November 2014 Guardians of Peace BBC / Reuters"
- Other sources: FBI statement of 19 December 2014, Wired, Fortune



Where Hollywood Meets Tomorrow

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VOL. 142 NO. 318 | Culver City, California | NOVEMBER 2014

HOLLYWOOD UNDER SIEGE: SONY PICTURES SERVERS WIPED IN MASSIVE HACK

Guardians of Peace Paralyze Studio Computers Ahead of 'The Interview'

By Daniel R. Ortega
Staff Writer

LOS ANGELES — Sony Pictures Entertainment confirmed late Monday that its computer systems were crippled in a major cyber attack, with hackers linked to the pro-Palestinian group "Guardians of Peace" claiming responsibility for wiping data from the studio's networks.

The attack, which reportedly began on Thursday, disrupted internal operations at the Culver City headquarters and affected email, employee files and production systems. Studio sources say the hackers left a digital message warning against the release of *The Interview*, the forthcoming comedy that has drawn controversy for its depiction of North Korean leader Kim Jong-un.

In a statement posted online, Guardians of Peace said the attack was a response to what it called "the studio's support for a disrespectful and dangerous film." The group threatened further action if the movie is released in theaters.

Sony has not officially commented on the full extent of the damage, but insiders say the hackers erased critical data and left a chilling image on several company computers: a red skeleton graphic with a message demanding the cancellation of the film.

The FBI is reportedly investigating the incident.

An abandoned workstation at Sony Pictures Entertainment's Culver City headquarters on Monday, after a reported cyberattack by the group Guardians of Peace. PHOTO BY REUTERS

FBI Joins Investigation Into Hack

Authorities probe link to North Korea, potential national security risks.

By David Lazarus
Staff Writer

LOS ANGELES — The FBI is investigating the massive cyberattack on Sony Pictures Entertainment, sources confirmed Monday, amid growing concerns that the breach may be linked to North Korea.

While the group Guardians of Peace has claimed responsibility, U.S. officials say the attack bears similarities to previous cyber operations attributed to North Korea, including the use of wiper malware and politically motivated messaging.

"This is a serious and ongoing investigation," said a senior U.S. law enforcement official. "We are working to determine the full scope of the breach and who is behind it."

The incident has raised new questions about the security of major media companies in the face of increasingly sophisticated cyber threats.

The Interview Faces Uncertain Future

Release plans in doubt after studio hack and threats.

By Jessica Roy
Entertainment Writer

LOS ANGELES — The future of *The Interview* remains uncertain after Sony Pictures was hit by a cyberattack that disrupted studio operations and led to the release of a threatening message from hackers.

The comedy, starring James Franco and Seth Rogen, was already at the center of global attention due to its portrayal of North Korean leader Kim Jong-un.

Continued on Page 4

Hollywood on High Alert

Security tightened across studios following cyberattack.

By Melissa Grant
Staff Writer

LOS ANGELES — In the wake of the Sony Pictures hack, major studios across Hollywood are reportedly increasing cybersecurity measures and reviewing their digital infrastructure.

Industry experts say the attack is a wake-up call for studios, many of which have been slow to adapt to the growing threat of cyber warfare.

"This isn't just about one movie or one studio," said cybersecurity analyst Robert Klein. "It's about the vulnerability of the entire entertainment industry."

Continued on Page 5

Support Grows for "The Interview"

Some see hack as an attack on free speech.

By Samantha Lee
Staff Writer

Despite the threats, supporters of The Interview are rallying behind the film, calling the hack an attack on freedom of expression.

"This is about more than a movie," said director Seth Rogen in a brief statement. "It's about the right to tell stories and not be silenced by bullies."

Social media has been flooded with messages of support, using the hashtag #IStandWithTheInterview.

Continued on Page 6

A City That Never Stops

Even in chaos, L.A. keeps its eye on the future.

Los Angeles remains resilient, no matter the challenge.

Page 12

Case 3: Bangladesh Bank Cyber Heist (Bangladesh, 2016)

Date / Timing	4 to 5 February 2016 (a Thursday night and Friday, when the bank's weekend began)
Victim	Bangladesh Bank, the central bank of Bangladesh, and its account at the Federal Reserve Bank of New York
Attack type	Malware on bank computers used to send fraudulent SWIFT payment messages
Amount	Fraudulent requests worth about $951 million; about $81 million actually reached the criminals

What happened

Hackers sent 35 fake payment instructions through the SWIFT system asking the New York Fed to transfer money from Bangladesh Bank's account. Five requests worth $101 million went through: $20 million to Sri Lanka and $81 million to accounts in the Philippines. The rest were blocked, partly because a spelling mistake ("fandation" instead of "foundation") made Deutsche Bank query one transfer. The malware also disrupted the bank's printer that normally printed SWIFT confirmations, which delayed the discovery.

Cause

- Attackers had gained access to the bank's network months before and installed malware that studied how SWIFT was used.
- The bank used cheap, unsecured network equipment and had weak security controls around its SWIFT terminals.
- Timing over a weekend and public holidays in both countries delayed responses.

Losses

- About $81 million was moved to the Philippines and quickly withdrawn through casinos and exchange agents.
- The governor of the central bank resigned after the incident.

Recovery and Cost

The $20 million sent to Sri Lanka was returned. Only a small part of the Philippines money, around $15 million, was recovered later; the bulk was never recovered. The Philippine central bank fined a bank involved (RCBC) about $21 million for its role in the money laundering. Bangladesh Bank also hired the cyber security firm FireEye (Mandiant) to investigate. The attack pattern was later linked by security researchers to the Lazarus Group.

Solution / Lessons

- SWIFT launched its Customer Security Programme (CSP) in 2016 with mandatory security controls for member banks.
- Banks now use stricter authentication, monitoring and isolated SWIFT terminals.
- Faster 24x7 response between central banks.

Source Evidence (Article / Photo)

- Suggested search: "Bangladesh Bank heist 2016 $81 million Reuters / BBC / The Daily Star"
- Other sources: Reuters investigation (July 2016), US Department of Justice statement on Lazarus Group (2018)



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THE FINANCIAL TRIBUNE

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VOL. XXXIX NO. 6,482 | MONDAY, FEBRUARY 15, 2016 | WORLD BUSINESS DAILY | $2.00

THE $81 MILLION CYBER HEIST: FRAUDULENT SWIFT TRANSFERS HIT BANGLADESH CENTRAL BANK

Typo Halts Billion-Dollar Loot Routed Through Global Accounts

Hackers breach central bank's SWIFT system in sophisticated attack

DHAKA — Hackers stole $81 million from Bangladesh Bank using fraudulent SWIFT transfer orders in one of the most brazen cyber heists in the history of global finance, bank officials and international investigators said, in an attack that could have been far larger had a typographical error not derailed some of the transactions.

The heist, carried out in early February, saw criminals send at least 35 fake payment instructions through the bank's SWIFT messaging system, seeking to transfer nearly $1 billion to accounts around the world. While most were flagged or rejected, five transactions totaling $81 million were successfully executed, primarily to accounts in the Philippines.

"It was a highly sophisticated and well-planned attack," said a senior Bangladesh Bank official, requesting anonymity. "Had the error not occurred in one of the key instructions, the losses could have reached close to a billion dollars."

The incident has raised fresh questions about cybersecurity at central banks and the vulnerability of the global financial messaging network.

Continued on Page 6

The Bangladesh Bank headquarters in Dhaka. Hackers exploited the bank's SWIFT access to send fraudulent transfer orders in early February 2016. PHOTO: REUTERS

Central banks worldwide on high alert

LONDON — Central banks across the globe have been placed on heightened alert following the Bangladesh Bank heist, with fears that the attack could signal a new wave of cyber threats targeting the international financial system.

Continued on Page 6

TIMELINE OF THE HEIST

Feb 4, 2016	Fraudulent SWIFT orders sent from Bangladesh Bank
Feb 5, 2016	$81 million transferred to accounts in the Philippines
Feb 6, 2016	Typo in one transfer halts $951 million payment
Feb 7, 2016	Bank detects irregularities
Feb 8, 2016	Bangladesh Bank notifies authorities
Feb 9, 2016	Global investigation underway

GLOBAL REACTION

"A wake-up call for the entire financial system."
— Mark Carney
Governor, Bank of England

"Cyber risk is now a core financial risk."
— Janet Yellen
Chair, U.S. Federal Reserve

"This changes the game for central bank security."
— Vítor Constâncio
Vice President, ECB

SWIFT UNDER SCRUTINY

BRUSSELS — The SWIFT cooperative, which provides the secure messaging network used by more than 11,000 financial institutions worldwide, said it is working with authorities to investigate the breach and review security protocols.

Continued on Page 7

GLOBAL MARKETS | DOW 15,660.18 ▼ -1.24% | FTSE 5,879.63 ▼ -0.86% | NIKKEI 16,085.45 ▲ +0.52% | DAX 9,205.17 ▼ -1.10% | GOLD 1,226.40 ▲ +0.33% | OIL 29.12 ▼ -2.47%

Case 4: Kudankulam Nuclear Power Plant Malware Attack (India, 2019)

Date / Timing	Malware detected in early September 2019; the plant operator confirmed the incident on 30 October 2019
Victim	Kudankulam Nuclear Power Plant (KKNPP), Tamil Nadu, operated by the Nuclear Power Corporation of India Limited (NPCIL)
Attack type	DTrack spyware / remote access malware
Attacker	Security researchers linked DTrack to the North Korea-associated Lazarus Group; this was not officially confirmed by India

What happened

News of a malware infection at the plant spread on social media in late October 2019, and NPCIL first denied it before admitting the next day that malware had been found in its administrative network. DTrack is designed to record keystrokes, browse files and transfer data to attackers. It was found on a computer used by an employee, in the administrative (office) network of the plant.

Cause

- A user in the administrative network was infected, most likely through a malicious file or phishing, and the malware spread to other systems.
- The plant's critical control systems are separated from the internet and the admin network ("air-gapped"), so they were not affected.

Losses

- NPCIL said that plant operations and safety were not affected.
- Some information on the administrative network may have been exposed, but the extent of data stolen has not been made public.
- Public concern about the safety of critical infrastructure grew.

Recovery and Cost

CERT-In informed NPCIL about the infection on 4 September 2019. Infected computers were isolated and cleaned, and a specialist team was set up to investigate and strengthen security. The cost of recovery has not been published.

Solution / Lessons

- Keep operational (control) networks separate from office networks.
- Regular security audits of critical infrastructure.
- Better reporting of cyber incidents and quick public communication.
- Monitoring of the admin network for spyware.

Source Evidence (Article / Photo)

- Suggested search: "Kudankulam nuclear plant malware NPCIL confirms DTrack October 2019 The Hindu / Indian Express / NDTV"
- Other sources: Kaspersky and Symantec reports on DTrack (2019), Reuters, NPCIL press statement of 30 October 2019



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VOL. LXXIII NO. 304 | THURSDAY, OCTOBER 31, 2019 | www.nationalchronicle.in | PRICE ₹6.00

NPCIL ADMITS MALWARE INFECTION AT KUDANKULAM NUCLEAR PLANT

DTrack Spyware Detected on Administrative Network; Operational Reactor Core Confirmed Safe and Air-Gapped

New Delhi/Chennai, Oct. 31

Nuclear Power Corporation of India Limited (NPCIL) has confirmed that malware, identified as DTrack, was detected on a section of the administrative network at the Kudankulam Nuclear Power Plant (KKNPP) in Tamil Nadu, officials said on Wednesday. The company emphasised that the plant's operational systems, including the reactor core control network, remain safe, secure and fully air-gapped from external and administrative networks.

In a statement, NPCIL said the infection was detected during a routine cybersecurity review last month. "The malware, known as DTrack, was found on certain administrative computers. There is no impact on safety, reactor operations or plant performance," the statement read.

NPCIL, a public sector under the Department of Atomic Energy (DAE), said the affected systems have been isolated, forensic analysis is underway, and additional safeguards are being implemented across all its facilities.

Security personnel on duty at the Kudankulam Nuclear Power Plant in Tamil Nadu. NPCIL has confirmed the presence of DTrack malware on its administrative network, asserting that all operational systems remain secure and isolated. — PTI

No Threat to Public Safety, Says NPCIL

Continued monitoring, additional countermeasures in place

By Our Staff Correspondent
New Delhi, Oct. 31

NPCIL officials reiterated that there is no threat to public safety or to the environment. "The reactor core and associated control systems are on a separate, air-gapped network and are not connected to the administrative IT infrastructure," a senior official said.

Cybersecurity experts said DTrack is a known espionage-focused malware that has targeted government and strategic organisations in the past. The breach, while contained, underscores the growing cyber risks faced by critical infrastructure in India.

Continued on Page 6

Centre Seeks Detailed Report

New Delhi: The Union government has sought a detailed report from NPCIL on the incident and the steps taken to prevent a recurrence, sources said. Page 6

A Growing Cyber Challenge

Experts warn that critical infrastructure must stay ahead of evolving threats in an increasingly complex digital landscape. Page 8

Kudankulam: Powering India's Future

The 2,000 MW plant is a key part of India's nuclear energy expansion, supplying reliable baseload power to southern states. Page 9

Case 5: Air India Passenger Data Breach (India, 2021)

Date / Timing	Attack on the SITA passenger system in February 2021; SITA disclosed it on 4 March 2021; Air India informed passengers on 21 May 2021
Victim	Air India, through its IT service provider SITA PSS (Passenger Service System); other airlines including Singapore Airlines and Malaysia Airlines were also affected
Attack type	Cyber attack on a third-party service provider leading to theft of passenger data
Data stolen	Personal data of about 4.5 million passengers worldwide registered between 26 August 2011 and 3 February 2021

What happened

Hackers broke into the servers of SITA, which stores and processes passenger information for many airlines. Data stolen included names, dates of birth, contact details, passport details, ticket information and frequent-flyer data. Air India said that credit card numbers and CVV/CVC codes were not stored with the provider, and that no passwords were exposed.

Cause

- The attack targeted the third-party provider and not Air India directly (a supply chain / vendor attack).
- Security firm Group-IB later published findings linking a similar campaign against the aviation industry to a state-linked group, but attribution is not officially confirmed.

Losses

- Personal data of millions of passengers exposed, creating risk of identity theft and phishing.
- Damage to the airline's reputation.
- The airline did not publish a financial loss figure.

Recovery and Cost

SITA secured the affected servers and cooperated with investigators, and Air India said it was working with external security experts. The airline asked passengers to change their frequent-flyer passwords, and it contacted credit card issuers where card data was involved. The cost of recovery has not been disclosed.

Solution / Lessons

- Check the security of third-party vendors that hold customer data.
- Limit how long and how much personal data is stored.
- Encrypt sensitive data such as passport numbers.
- Inform customers quickly after a breach.

Source Evidence (Article / Photo)

- Suggested search: "Air India data breach SITA 4.5 million passengers May 2021 Reuters / Economic Times / Hindustan Times"
- Other sources: SITA statement (March 2021), Air India notice to passengers (21 May 2021), Group-IB report on aviation attacks



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DATA OF 4.5 MILLION AIR INDIA PASSENGERS LEAKED IN MAJOR SITA SERVER CYBER ATTACK

Passports, Ticket Records, and Flyer Details Stolen in Global Third-Party Aviation Breach

By Ananya Deshmukh and Reuters
NEW DELHI, May 22, 2021

The personal data of around 4.5 million Air India passengers has been compromised in a large-scale cyber attack targeting SITA, a global aviation technology provider, the airline and sources familiar with the matter said on Friday.

The breach, which is believed to have occurred earlier this month, involved unauthorised access to data stored on SITA's servers, affecting multiple airlines worldwide. Air India confirmed that passenger information including names, passport details, ticket records, frequent flyer details and contact information may have been stolen.

"We take the matter very seriously and are working closely with SITA to understand the full scope and impact of the incident," Air India said in a statement.

SITA, which provides IT solutions to over 400 airlines and airports around the world, said the attack affected some of its systems and that an investigation is ongoing. "We have contained the incident and are implementing additional security measures," SITA said in a statement.

Passengers at Indira Gandhi International Airport in New Delhi on Friday. Air India confirmed that data of 4.5 million passengers may have been compromised in a cyber attack on SITA, a global aviation IT provider. AFP

Aviation cybersecurity experts said the breach underscores the growing risks faced by airlines due to their reliance on third-party service providers. "This incident highlights the need for greater visibility and stronger data protection across the aviation ecosystem," said one industry analyst.

Air India has advised affected passengers to remain vigilant against potential phishing attempts and unauthorised use of their personal

The Indian government said it is in touch with the airline and SITA and is monitoring the situation. More details are expected as the investigation progresses.

▶ Continued on Page 6

Conclusion: Common Lessons

- Most of these attacks began with a simple entry point: a phishing email, a stolen vendor password or an infected office computer.
- Poor network separation let attackers move from a small foothold to critical systems (Target payment network, Sony internal network, Bangladesh Bank SWIFT terminals).
- Third-party vendors are a major risk (Target's HVAC contractor, SITA for Air India), so their security must also be checked.
- Critical services such as banks, retailers and power plants need incident response plans and regular drills.
- Employee awareness and strong access control reduce the chance of an initial break-in.

References

- BBC News, Reuters, Wired, The Hindu, Times of India, Economic Times, NDTV, Hindustan Times (news coverage of each case)
- US Senate Commerce Committee: A Kill Chain Analysis of the 2013 Target Data Breach (2014)
- FBI statement on the Sony Pictures hack (19 December 2014)
- Reuters investigation on the Bangladesh Bank heist (2016) and SWIFT Customer Security Programme
- NPCIL press statement (30 October 2019) and Kaspersky / Symantec reports on DTrack
- SITA and Air India notices on the passenger data breach (2021)